Project 99: Integrated E-Commerce Order Processing and Reconciliation

Python Mini Project – College Assignment

1. Aim

To manage e-commerce product orders, discounts, dispatches, returns and stock updates; calculate net revenue, identify discrepancies and produce a final inventory summary.

2. Algorithm

1. Start the program.
2. Store product names, prices, stock, order quantities, dispatch quantities, returns and discounts.
3. For each product, check available stock against dispatch quantity.
4. Calculate discount, selling price, sales, refund and net sales.
5. Update inventory after dispatch and return.
6. Compare order quantities with dispatch quantities and report differences.
7. Display remaining inventory, refunds, revenue and total discrepancy.
8. Stop.

3. Complete Python Program

```
# Project 99: Integrated E-Commerce Order Processing

print("=======================================")
print(" E-COMMERCE ORDER PROCESSING SYSTEM")
print("=======================================")

products = ["Laptop", "Mouse", "Keyboard"]
prices = [50000, 500, 1000]
stock = [10, 30, 20]
orders = [5, 12, 8]
dispatch = [4, 12, 7]
returns = [1, 2, 1]
discounts = (10, 5, 8)

total_revenue = 0
total_refund = 0
total_discrepancy = 0

print("\nORDER PROCESSING REPORT")

for i in range(len(products)):
    print("\n---------------------------------")
    print("Product:", products[i])
    print("Price:", prices[i])
    print("Initial Stock:", stock[i])
    print("Ordered:", orders[i])
    print("Dispatched:", dispatch[i])
    print("Returned:", returns[i])

    if dispatch[i] <= stock[i]:
        discount_amount = prices[i] * discounts[i] / 100
        selling_price = prices[i] - discount_amount
        sales = dispatch[i] * selling_price
        refund = returns[i] * selling_price
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        stock[i] = stock[i] - dispatch[i] + returns[i]
        net_sales = sales - refund
        total_revenue += net_sales
        total_refund += refund

        print("Discount:", discounts[i], "%")
        print("Selling Price:", selling_price)
        print("Sales Amount:", sales)
        print("Refund Amount:", refund)
        print("Net Sales:", net_sales)
        print("Updated Stock:", stock[i])

        if orders[i] != dispatch[i]:
            difference = orders[i] - dispatch[i]
            total_discrepancy += difference
            print("Order Discrepancy:", difference)
        else:
            print("Order Status: Matched")

        if returns[i] > dispatch[i]:
            print("Return Discrepancy Detected")
    else:
        print("Insufficient Stock for Dispatch")

print("\n======================================")
print("       FINAL INVENTORY SUMMARY")
print("======================================")
for i in range(len(products)):
    print(products[i], "- Remaining Stock:", stock[i])
print("\nTotal Refund:", total_refund)
print("Total Net Revenue:", total_revenue)
print("Total Order Discrepancy:", total_discrepancy)
print("\n======================================")
print("       RECONCILIATION COMPLETED")
print("======================================")
```

4. Actual Sample Output

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 E-COMMERCE ORDER PROCESSING SYSTEM
======================================

ORDER PROCESSING REPORT

---------------------------------
Product: Laptop
Price: 50000
Initial Stock: 10
Ordered: 5
Dispatched: 4
Returned: 1
Discount: 10 %
Selling Price: 45000.0
Sales Amount: 180000.0
Refund Amount: 45000.0
Net Sales: 135000.0
Updated Stock: 7
Order Discrepancy: 1

---------------------------------
Product: Mouse
Price: 500
Initial Stock: 30
Ordered: 12
Dispatched: 12
Returned: 2
Discount: 5 %
Selling Price: 475.0
Sales Amount: 5700.0
Refund Amount: 950.0
Net Sales: 4750.0
Updated Stock: 20
```

```
Order Status: Matched

---------------------------------
Product: Keyboard
Price: 1000
Initial Stock: 20
Ordered: 8
Dispatched: 7
Returned: 1
Discount: 8 %
Selling Price: 920.0
Sales Amount: 6440.0
Refund Amount: 920.0
Net Sales: 5520.0
Updated Stock: 14
Order Discrepancy: 1

====================================
       FINAL INVENTORY SUMMARY
====================================
Laptop - Remaining Stock: 7
Mouse - Remaining Stock: 20
Keyboard - Remaining Stock: 14

Total Refund: 46870.0
Total Net Revenue: 145270.0
Total Order Discrepancy: 2

====================================
       RECONCILIATION COMPLETED
====================================
```

5. Flowchart (Text Form)

START

↓ Store product and order details

↓ For each product: Is dispatch within available stock?

YES → Apply discount → Calculate sales and refund → Update stock

NO → Report insufficient stock

↓ Compare orders with dispatch quantities

↓ Display inventory and revenue summary

STOP

6. Formulas

Discount = Price × Discount Percentage / 100

Selling Price = Price – Discount

Net Revenue = Sales Amount – Refund

Updated Stock = Initial Stock – Dispatched + Returned

7. Python Concepts Used

Variables, data types, operators, conditional statements, loops, strings, lists and tuples. The program uses only basic Python syllabus concepts.

8. Final Inventory Summary

Product	Initial Stock	Remaining Stock
Laptop	10	7
Mouse	30	20
Keyboard	20	14
TOTAL	**60**	**41**

Total Refund: Rs. 46,870 | Net Revenue: Rs. 1,45,270 | Order-Dispatch Discrepancy: 2 units

9. Result

The program successfully processes the sample orders, discounts and returns, and reports remaining stock, net revenue and discrepancies. The sample assumes returned units are valid and restockable; a production system should validate returns before issuing refunds or changing stock.